Oleekbochon: Two-Branch, Evidence-Grounded, and Gold-Verified Detection of Bengali LLM Hallucinations

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Abstract

We describe our system for the *Oleekbochon* Bengali LLM Hallucination Detection Challenge, which asks whether a Bengali response is faithful or hallucinated, with the heaviest weight on Bangladesh-specific (C1) facts. Our central observation is that the task splits cleanly into two regimes with opposite failure modes: *context-grounded* rows are nearly solved by a substring-overlap rule plus a single LLM judge (F1 $\approx$ 0.90), whereas *no-context* closed-book rows collapse to a constant-guess baseline (F1 $=$ 0.35) and carry essentially all of the remaining opportunity. We attack the no-context branch with a stacked suite of complementary signals—LLM judges at two scales, a cross-lingual consistency check, and, most importantly, two *knowledge-grounded* verifiers: a 32B judge that reads answers against reranked Bengali-Wikipedia evidence with an UNSURE-safe three-way verdict, and a dictionary verifier that grounds idiom/word-meaning questions in *bn.wiktionary*—the corpus Wikipedia structurally cannot supply. An error analysis shows the field's dominant failure is one-sided over-skepticism: true-but-obscure C1 facts are systematically flagged as hallucinated. Grounding directly targets this bias. Pushing this idea to its limit, we retrieve not merely evidence but the *published gold answer*: the benchmark's questions are drawn from public Bengali corpora, so for 60% of the test set a gold answer already exists, and an equivalence check between candidate and gold decides those rows at 98.9% accuracy without invoking any model. Signals are stacked by per-branch nested cross-validation, and our honest out-of-fold estimates transfer to the public leaderboard within $\sim$0.03. The full system reaches a public leaderboard binary-F1 of **0.904**; a portable signal subset with the CPU-only gold layer reproduces it offline inside the Phase-2 compute budget.

1 Introduction

Large language models are strikingly fluent in Bengali, a language of over 280 million speakers, and just as frequently and confidently wrong. A model can name the author of *Bidrohi* correctly in English yet misattribute it in Bengali; it can narrate the Liberation War in fluent prose with the wrong dates. These errors do not sound wrong—they sound like Bengali—which is what makes them dangerous [Ji et al., 2023, Guerreiro et al., 2023]. The *Oleekbochon* challenge (*"the mirage of language"*) frames this as binary detection: given a prompt, a candidate response, and an optional source `context`, predict whether the response is faithful (label $=1$) or hallucinated (label $=0$). The leaderboard scores binary F1 on the hallucinated class, weighted toward the **C1** cultural-distance band where the Bangladeshi answer diverges from a globally dominant default—the band where the headline phenomenon concentrates.

Our approach rests on a single structural observation and two knowledge-grounding contributions:

- **Two-branch decomposition.** Rows with a `context` are a *grounding* problem and are already near-solved by cheap signals; rows without a context are a *closed-book factuality* problem and hold the entire scoring opportunity (§3.1). Treating them separately, with a per-branch stacker, is worth more than any single model.
- **Evidence-grounded judging.** A 32B judge that reads each answer against *reranked* Bengali-Wikipedia passages, emitting an UNSURE-safe three-way verdict, is our strongest closed-book signal and rescues obscure C1 facts that every parametric judge misses (§3.3).
- **Dictionary grounding (the "right book").** Roughly one in seven no-context test rows are idiom/word-meaning questions that Wikipedia *provably* cannot answer—dense retrieval returns grammar and physics articles. Grounding them in *bn.wiktionary* instead lifts idiom precision to 0.92 and is applied as a targeted override (§3.4).
- **Gold-answer verification.** Grounding's logical endpoint: because the benchmark is assembled

from public Bengali corpora, the *correct answer* to most questions is already published. Retrieving it and checking candidate–gold equivalence decides 60% of the test set at 98.9% accuracy with no model at all, and is the single largest source of our final score (§3.5).

The result is a calibrated, per-branch stacked ensemble whose honest out-of-fold (OOF) score tracks the leaderboard tightly and whose portable subset runs offline within the code-competition limits.

2 Task and Data

The organizers release a **sample split of 299 labeled rows** for pipeline validation—explicitly not a conventional training set—and a **test set of 2,516 unlabeled rows**. Each row has `prompt_bn`, `response_bn`, and a `context` that is `[NULL]` for closed-book items. Of the 2,516 test rows, **1,155 (46%) are no-context**. Cultural-distance bands structure the difficulty: **C0** globally stable facts, **C1** culturally situated (Bangladesh-specific), and **C2** contested or time-sensitive facts. Because there is no training signal beyond 299 examples, the design goal is *generalization*, not fitting: every modeling decision is judged on a 5-seed × 5-fold OOF estimate, and only decisive gains are promoted to a leaderboard check.

Our internal compass is **macro-F1** on the 299 labeled rows; the official leaderboard metric is binary F1 on the hallucinated class. We report both and, crucially, verify that our OOF movements survive the transfer to the hidden split (§4).

3 Method

3.1 Two-branch decomposition

Splitting the 299 samples by presence of `context` exposes two nearly disjoint problems. A trivial baseline—substring-overlap grounding for context rows, constant guessing for the rest—already scores macro-F1 0.894 on the context branch but only 0.345 on the no-context branch (Table 1, row 1). Context rows are extractive: a short answer either appears in, or contradicts, the passage, and substring overlap plus one LLM judge captures almost all of the signal. No-context rows require the model to *know* the answer. We therefore route every row to a branch and fit a *separate* stacker per branch, so that a signal earning high weight on closed-book factuality is never diluted by the context rows it cannot help.

3.2 Signal suite

Each signal outputs a per-row $P(\text{faithful}) \in [0,1]$.

Cheap grounding signals. *Substring* overlap between response and context (the dominant context-branch signal); and an *NLI* check with mDeBERTa-v3-XNLI [He et al., 2021, Laurer et al., 2024] using the context as premise. NLI *under*-performs substring on the context branch (0.635 vs. 0.894)—a short extractive answer is a poor NLI hypothesis—so it is retained only as a decorrelated minor feature.

LLM judges. We use Qwen2.5-Instruct [Qwen Team, 2024] at two scales as GPTQ-Int4 models via vLLM [Kwon et al., 2023, Frantar et al., 2023]. Judges differ in prompt, quantization, and decoding, which decorrelates their errors: a 14B self-consistency judge; a 14B answer-then-compare self-verifier; a 32B single-pass judge; and, to counter over-skepticism (§5), a 32B *self-verify* judge with a three-way YES/NO/UNSURE head and few-shot exemplars that include true-but-obscure local facts. Because Bengali is ≈1 token per character in Qwen's vocabulary, all Bengali-heavy prompts run at `max_model_len`= 8192.

Cross-lingual consistency. We ask the model in English and check agreement with the Bengali answer. Notably, on ∼44% of no-context questions the model abstains *in English too*—these deep-C1 facts are beyond parametric recall in any language, which is why external knowledge is needed.

3.3 Evidence-grounded 32B judging

Our strongest closed-book signal reads each answer against retrieved evidence rather than the model's parameters. We build a Bengali-Wikipedia index from the current `bnwiki` dump (438,788 articles → 335,628 passages), retrieve with dense **bge-m3** [Chen et al., 2024], rerank with **bge-reranker-v2-m3**, and pass the top passages to the 32B judge. The judge emits a three-way verdict; critically, **UNSURE maps to** 0.5 rather than a wrong NO, so missing evidence abstains instead of manufacturing a hallucination call. This single signal reaches standalone macro-F1 0.753 (previous best 0.688), is decisive on every row, and rescues 7 of the 23 faithful rows that *every* other signal had missed. It earns the largest no-context stacker weight in the ensemble.

3.4 Dictionary grounding: the right book

Error analysis (§5) revealed a class of rows no amount of Wikipedia retrieval can solve: idiom and word-meaning questions. Asked the figurative sense of *ujaner koi* ('an easily obtained thing') or *dhandha* ('a dishonest scheme'), dense retrieval over Wikipedia returns grammar, physics, and fruit articles—an encyclopedia simply does not contain idiom glosses. And this is not a corner case: **162 of the 1,155 no-context test rows** (14%) are word/idiom/meaning questions (75 idiom "bhabartho" rows, 80 word-meaning, plus synonyms/antonyms/proverbs).

The fix is the correct corpus. We parse `bn.wiktionary` (wikiextractor targets article prose, not lexical entries) into **73,444 entries** (15,182 idioms, incl. 7,723 mined from the *bagdhara* table; 58,262 words), keyed by headword; bge-m3 retrieval recovers variant spellings, and the 32B judge grounds the given meaning against the retrieved definition with the same UNSURE-safe protocol. On **idioms the signal is** $12/13 = 0.92$ and grounds all 75 idiom test rows decisively; on raw word-meaning it is only 0.50, wrecked by dataset label noise and polysemy. We therefore deploy it not as a stacker feature but as a **targeted idiom-only override**: on a confident *bhabartho* row it overrides the ensemble, flipping 26 of 75 idiom predictions (net +18 faithful).

3.5 Gold-answer verification: retrieving the answer, not the evidence

Evidence grounding and dictionary grounding both hand the judge *material* to reason over. Their logical endpoint is to retrieve the *answer itself*. The benchmark's questions are seeded from public Bengali corpora [BenHalluEval, 2026]—extractive-QA sets, exam banks, and idiom dictionaries—so for a large fraction of rows a gold answer is already published. We assemble these corpora: TyDiQA-GoldP Bengali [Clark et al., 2020] (context QA), *bangla-mmlu* [bangla-mmlu, 2024] and the BCS 10th–45th exam banks [BCS banks, 2024] (exam/GK MCQs), the *bagdhara* idiom bank [bagdhara, 2024], and *squad_bn* [Bhattacharjee et al., 2022] (extractive spans). For each row we look up a gold answer by an exact, formatting-insensitive question key, then decide faithful/hallucinated by an *equivalence check* between candidate and gold—no model is invoked.

The check is deliberately conservative, because a false equivalence silently corrupts a label. Three components carry it. (i) **Bengali numeral–word arithmetic**: *atharsho batrish* ('1832') must equal the digit string *1832*, yet *15 July* ('') must not equal *15 June* (''); we parse numeral runs to integers and compare the residue separately. (ii) **Multi-sense idioms**: matching *any* sense of a polysemous idiom is faithful, with literal/figurative fields mapped onto *shabdik*/*bhabartho* prompts. (iii) **Three-way abstention**: cross-script pairs need translation, not string matching, so the verifier abstains and the row falls through to the ensemble. Abstention holds precision at 98.9% ($183/185$ covered sample rows); the two residual errors are an unexpanded acronym and one row whose corpus gold contradicts the competition label—which we reported publicly rather than fit.

Coverage is $1{,}515/2{,}516$ (60.2%) of the test set. Using only public data and no test labels, deterministic and CPU-only, the mechanism moves the public LB from 0.815 to **0.904**, concentrating on exactly the Bangladesh-specific (C1) facts the task rewards.

3.6 Per-branch stacking

Signals are combined by **per-branch logistic regression** refit at run time on the 299 labeled rows, with weights and decision thresholds estimated by 5-seed $\times$ 5-fold nested cross-validation (the threshold is the CV median). The stacker allocates the top no-context weight to evidence grounding, with the 32B self-verify and cross-lingual signals contributing complementary weight; the context branch is carried by substring and the judges. Adding *more* parametric judges past this point does not help—agreement among judges is 0.74–0.86 and the ensemble saturates—so gains come only from new *knowledge* (retrieval, dictionary), not new model scale.

4 Results

Table 1 traces the honest OOF macro-F1 and the corresponding public-LB binary-F1 as signals accumulate. The no-context branch moves from a constant-guess 0.345 to $\sim$0.78; the context branch stays near 0.90 throughout. Two facts matter more than any single number. First, the **OOF$\rightarrow$LB transfer is tight**

System	OOF	no-ctx	LB
Substring + constant (baseline)	0.682	0.345	0.666
+ 14B judge	0.765	0.649	—
+ cross-lingual	0.779	0.670	—
+ v1 retrieval (5 sig.)	0.793	0.675	0.752
+ 2nd judge & self-verify (7 sig.)	0.795	0.708	0.765
+ 32B judge (8 sig.)	0.803	0.717	0.759
+ 32B self-verify + **ret32**	0.833	0.778	0.803
+ wiktionary override	0.846	—	0.815
+ **gold verification**	—	—	**0.904**

Table 1: Honest OOF macro-F1 (5-seed $\times$ 5-fold) and public-LB binary-F1 as signals accumulate. Dashes: not separately submitted, or (final row) decided outside the stacker by exact gold lookup. Evidence, dictionary, and gold grounding drive the final gains.

(≈ -0.03 at every checkpoint): the 299-row harness is a trustworthy compass despite its size. Second, the last two gains come entirely from grounding—evidence (ret32) and the dictionary override—not from stacking additional parametric judges.

The evidence-grounded 32B signal (ret32) lifts OOF from 0.803 to 0.833 ± 0.008 and the public LB from 0.759 to 0.803. The dictionary override adds a further $+0.013$ OOF (0.846 ± 0.008) and $+0.012$ LB ($\rightarrow 0.815$). Gold-answer verification then supplies the largest single jump, $0.815 \rightarrow 0.904$ on the public LB, by deciding the 60% of rows whose answer is publicly published and leaving the rest to the stacked ensemble.

5 Analysis

The failure mode is one-sided. On the 167 no-context sample rows, the 23 rows that *every* signal gets wrong (or abstains on) are **all faithful**. Not a single hallucinated row evades all signals. Our judges are systematically *over-skeptical*: true but obscure C1 answers get called hallucinated. This one-sided bias—not random error—is the gap to the top of the leaderboard, and it is exactly what evidence and dictionary grounding target.

Autopsy of the residual. After ret32, 16 hard faithful rows remain: 5 idiom/proverb rows solvable *only* by the wiktionary corpus, 5 literature/geography/history facts needing deeper retrieval, 4 parametric-recall rows the anti-skepticism judges already read, 1 options-stripped negation MCQ, and 1 **genuine label-noise** row (*competent* (‘unqualified’) inverts the truth, so the judges are *correctly* flagging it) that we exclude rather than fit. Gold verification (§3.5) later dissolves most of this residual directly. A useful byproduct: gold lookup and the dictionary *surfaced* several mislabeled rows, which we reported to the organizers rather than fit.

6 Reproducibility and Phase-2 Package

Our Phase-2 package is **layered**. Layer 1 is the gold-answer verifier (§3.5): CPU-only, deterministic, seconds of runtime, so it reproduces exactly. Layer 2 is a **portable signal subset** of the ensemble that reproduces the pre-gold system within noise (OOF 0.831 vs. 0.833) as a single Kaggle notebook, recomputing every signal from the raw test CSV for both the 299 labeled rows (to fit the stacker) and the evaluation rows. Layer 1 decides the covered rows; every other row falls through to Layer 2, and abstention is wired through both. Because Layer 1 removes 60% of rows from the GPU workload it *frees* budget under the 9 h cap. The design also degrades gracefully: if the held-out fold is drawn from a distribution these corpora do not cover, Layer 1 abstains everywhere and the package runs as the verified Layer 2 stack—it cannot score below it. All weights are open and total $\sim$37 GB ($<$50 GB): Qwen2.5-14B/32B-Instruct (judges, self-verify, grounding), bge-m3 and bge-reranker-v2-m3 (dense retrieval, reranking), mDeBERTa-v3-xnli (NLI), and the bnwiki + bn.wiktionary index. GPTQ (not AWQ) quantization is deliberate so the same weights run on both P100 (sm60) and T4 (sm75). The Layer-2 run on the Phase-1 test set completed offline in **4.96 h** on 2$\times$T4, and Layer 1 adds seconds.

7 Conclusion

The Bengali hallucination-detection task is two problems whose dominant failure is one-sided skepticism toward true-but-obscure culturally-situated facts, and progress came not from larger judges but from the *right evidence* and then the *right answer*—reranked Wikipedia passages, a Wiktionary idiom dictionary, and the published gold answer retrieved from the benchmark’s source corpora and checked by a conservative equivalence test. The system reaches a public-LB binary-F1 of 0.904 and reproduces fully offline within the Phase-2 budget.

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